Reactive Autonomous Mapping: Implementation of Follow-The-Gap and SLAM Mapping on an Ackermann-Steered RC Platform Using ROS2

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***Abstract*—This paper presents a reactive autonomous mapping framework for a 1/10-scale Ackermann-steered RC platform operating under ROS2. The work addresses the bootstrapping problem in mobile robotics: map-based planners require an occupancy-grid map, but building one safely requires navigation through an initially unknown space. The proposed system combines LiDAR-based Follow-the-Gap (FTG) navigation with SLAM using `slam_toolbox`, allowing the vehicle to explore a structured indoor environment with static obstacles without a prior map, global planner, predefined trajectory, or external intervention. Experiments on a 15 m $\times$ 4 m test track showed that the vehicle traveled 38.6 m with zero emergency braking activations and produced a geometrically coherent occupancy-grid map suitable for subsequent map-based navigation. These results demonstrate low-cost autonomous map acquisition under real sensing constraints.**

***Index Terms*—SLAM mapping, Follow-the-Gap, autonomous navigation, ROS2**

I. Introduction

Autonomous ground vehicles have become an active area of research in mobile robotics due to their growing applicability in logistics, warehouse automation, infrastructure inspection, precision agriculture, search and rescue operations, and intelligent transportation systems. Advances in sensing technologies, computational capabilities, and middleware have enabled the development of increasingly sophisticated autonomous navigation frameworks capable of operating in complex and dynamic environments. These systems typically integrate perception, localization, planning, and control modules to achieve safe and reliable autonomous operation.

In recent decades, numerous navigation strategies have been developed, which can generally be classified into reactive and map-based approaches, depending on the information used for decision-making.

Reactive navigation methods generate control actions directly from real-time sensor observations, without requiring prior knowledge of the environment. These approaches are particularly appealing in unknown or partially known scenarios due to their low computational complexity, rapid response to environmental changes, and ability to operate using only information from onboard sensors. Classic reactive strategies include wall-following, obstacle avoidance, and Follow-The-Gap (FTG) algorithms, which identify navigable regions within LiDAR observations and guide the vehicle toward the safest available path [1]. Thanks to their simplicity, robustness, and real-time applicability, reactive methods have been widely adopted in robotic exploration, obstacle avoidance, and the autonomous discovery of unknown environments.

In contrast, map-based navigation methods rely on a pre-existing representation of the environment to generate collision-free trajectories toward a desired target. These systems typically combine localization algorithms, global route planning, and trajectory tracking to achieve efficient navigation in structured environments [2]. Since navigation decisions are made using global environmental information, map-based methods generally provide route optimization, smoother trajectories, and greater navigation efficiency compared to purely reactive approaches.

However, the successful deployment of map-based navigation poses a fundamental challenge in autonomous robotics. Before a robot can use global planning algorithms, it must have a reliable map of the environment in which it is to operate. At the same time, generating such a map requires the robot to navigate safely through an initially unknown workspace. In many practical applications, maps are generated through manual teleoperation, predefined exploration routines, or human-assisted data collection processes. While effective,

these approaches reduce the robotic system's autonomy at its most critical stage and increase operational complexity, especially in environments that are inaccessible, unsafe for teleoperation, or frequently changing. Therefore, relying only on vehicle-external operations for mapping exposes the platform to severe collision risks if the control link is interrupted or becomes unavailable before a base map is established.

Reactive navigation offers a promising alternative for addressing this limitation. Since they do not require prior global representations or remote commands, these methods can be deployed immediately in completely unknown environments [3]. When integrated with Simultaneous Localization and Mapping (SLAM) techniques—such as modern ROS2 and LiDAR-based frameworks—reactive obstacle avoidance provides continuous environmental coverage while constructing a reliable grid-based representation of the workspace's occupancy [4], [5]. As a result, this approach enables the generation of reliable occupancy-grid maps, which serve as the basis for the subsequent deployment of global planning and map-based navigation architectures.

This idea is closely related to the concept of Active SLAM (ASLAM), in which a robot autonomously selects its path while simultaneously constructing a representation of the environment [6]. Instead of relying on predefined trajectories or human teleoperation, active exploration strategies aim to maximize environmental coverage and information acquisition during the mapping process. Reactive navigation methods such as Follow-The-Gap can, for example, be configured to direct the robot toward the boundary between known and unknown space, enabling the progressive discovery of new regions and improving map completeness [6]. In this context, reactive navigation methods can be interpreted as lightweight exploration mechanisms that naturally guide the robot toward navigable free-space regions while continuously providing new observations to the SLAM framework [7]. Furthermore, integrating accurate odometry and IMU data significantly reduces localization error during this mapping phase, ensuring high-fidelity grid representations [8].

From an operational perspective, reactive navigation can contribute to autonomous mapping in two complementary ways. First, during the initial deployment phase, when no map of the environment is available, reactive obstacle avoidance allows the robot to safely navigate through unknown regions and acquire the observations necessary for map construction [9]. This capability is particularly crucial in scenarios where communication limitations or terrain complexity make human teleoperation impossible. Second, once a preliminary map has been generated, reactive behaviors remain valuable as a local corrective layer capable of responding to unexpected obstacles, environmental changes, or inconsistencies not represented in the existing map [7], [10]. Therefore, reactive navigation should not be considered solely as a mechanism for obstacle avoidance, but also as a technology capable of both map generation and the maintenance of adaptive maps in changing environments [9]. This work focuses on the first of these stages, addressing the problem of autonomous map acquisition in a previously unknown environment using reactive navigation.

Consequently, rather than viewing reactive and map-based navigation as contradictory paradigms, they should be interpreted as complementary and sequential stages within a comprehensive and autonomous exploration workflow. From this complementary perspective, sensor-based reactive navigation acts as a facilitating mechanism for initial deployment, mitigating operational risks in unfamiliar spaces to safely extract and consolidate the environmental representation required by subsequent high-level map-based frameworks.

This article presents an autonomous mapping framework for a 1/10 scale Ackermann-steering robotic vehicle, based on the integration of the Follow-The-Gap and SLAM reactive navigation algorithms within a ROS2 architecture. Unlike conventional map-based navigation systems, the proposed framework performs exploration using only LiDAR observations, simultaneously generating a grid map of the environment's occupancy. The aim of this work is to evaluate whether a safe and computationally efficient reactive navigation strategy can autonomously explore a closed environment with obstacles and produce a map of sufficient quality to support the future implementation of map-based navigation algorithms.

Experimental validation is carried out on an approximately 15 m $\times$ 4 m indoor physical test track with multiple static obstacles. The performance of the proposed system is evaluated using quantitative navigation metrics, including distance traveled, forward speed, angular velocity, and safety braking interventions, as well as through a qualitative evaluation of the generated occupancy grid map and the estimated vehicle trajectory, and a visual analysis in terms of coherence and geometric consistency with respect to the test track.

The remainder of the article is organized as follows: Section II presents the proposed methodology, including the Follow-The-Gap navigation strategy, the Ackermann experimental platform, the odometry estimation framework, the SLAM-based mapping architecture implemented in ROS2, and a brief description of the experimental environment. Section III presents the experimental results and analyzes both the navigation performance and the quality of the generated map. Finally, Section IV summarizes the main findings and outlines future research directions.

II. Methodology

A. Follow-The-Gap Reactive Navigation

The autonomous navigation strategy is based on the Follow-The-Gap (FTG) method, originally proposed by [11]. FTG is a reactive obstacle-avoidance algorithm that generates steering commands from LiDAR observations without requiring a pre-existing map or global trajectory. This characteristic makes it suitable for local navigation in unknown environments while a separate Simultaneous Localization and Mapping (SLAM) process constructs an environmental representation [12].

The control loop operates at a fixed frequency of 20 Hz. At each control cycle, the LiDAR scan is processed to identify obstacle-free angular sectors, referred to as *gaps*. Candidate

gaps are defined by contiguous sequences of LiDAR beams that satisfy the implemented admissibility conditions. Before gap selection, a Time-to-Collision (TTC)-based filtering stage evaluates the potential collision risk associated with the observed directions using the vehicle's longitudinal velocity, $v(t)$, estimated from the incremental wheel encoder. Beams that do not satisfy the configured TTC criterion are excluded from the candidate gaps. The gap with the largest angular width is then selected.

The desired heading, $\theta_d(t)$, is defined as the angular midpoint of the selected gap. The heading error, $\theta_e(t)$, represents the angular deviation between the desired heading and the vehicle heading. A proportional controller generates the steering command according to

$$\delta(t) = K_p \theta_e(t), \tag{1}$$

where K_p is the proportional gain, set to $K_p = 1.0$, and $\delta(t)$ is the commanded steering angle, expressed in degrees at the actuator interface. The heading-error representation and controller implementation determine the units associated with the proportional gain.

Under normal navigation conditions, the vehicle operates at a nominal forward velocity of 0.5 m/s. An emergency intervention layer monitors a forward LiDAR sector with a total angular width of 18°. The intervention is triggered if the estimated TTC falls below 0.85 s or if an obstacle enters the configured safety-distance threshold of 1.2 m. When an intervention is triggered, the linear velocity command is overridden and a steering command of $+0.9^\circ$ or -0.9° is applied according to the detected obstacle orientation. This layer is designed to override nominal navigation commands when the specified intervention conditions are met; its configuration alone does not establish a general guarantee of collision avoidance.

B. Experimental Platform and Hardware

The experimental platform is a commercial 1/10-scale Ackermann-steered RC vehicle, the AMORIL RTR Brushless Fast RC Off-Road Monster Truck, shown in Fig. 2. The vehicle retains its original drivetrain, electronic speed controller (ESC), and steering servo. Additional sensing, computation, and communication modules were integrated to support autonomous navigation. The complete platform, including its onboard electronics and power systems, has an approximate total mass of 2.82 kg and a wheelbase of approximately 0.31 m.

The vehicle uses an Ackermann steering configuration, with steering commands applied to the front axle and propulsion provided by the rear drivetrain. Steering and throttle commands are transmitted to the Yahboom expansion board, which generates the pulse-width modulation (PWM) signals required by the steering servo and ESC.

A Raspberry Pi 5 runs ROS2 Jazzy Jalisco nodes for sensor acquisition, data communication, and low-level vehicle interfacing. The onboard LiDAR connects to the Raspberry Pi through USB. An ESP32 acquires pulses from an incremental encoder mechanically coupled to a rear wheel and transmits the encoder data to the Raspberry Pi through a USB serial connection. The Raspberry Pi also communicates with the Yahboom ROS Expansion Board V3.0, which provides inertial measurements and interfaces with the steering and propulsion actuators.

Environmental perception is provided by a centrally mounted SLAMTEC RPLIDAR A1M8 two-dimensional LiDAR. The sensor provides 360° planar scanning over an operational range of approximately 0.15–12 m, with a typical scan frequency of approximately 5.5 Hz. Its measurements are used for reactive obstacle avoidance and online environment mapping.

The inertial sensing subsystem uses the Yahboom ROS Expansion Board V3.0, which provides accelerometer, gyroscope, and magnetometer measurements. A Madgwick attitude filter processes the inertial measurements using a gain of $\beta = 10^{-4}$ and a fixed sampling period of 0.1 s (10 Hz). The incremental encoder provides the wheel-rotation measurements used to estimate longitudinal velocity. The estimated velocity and IMU-based heading are then used to calculate the planar dead-reckoning position supplied to the SLAM framework.

The system architecture is presented in Fig. 1. Computation is distributed between the onboard Raspberry Pi and an external laptop. The Raspberry Pi handles sensor acquisition, inertial processing, encoder communication, and actuator interfacing. The laptop runs the high-level ROS2 processes, including FTG navigation, emergency intervention, and SLAM. Both computers communicate over Wi-Fi using CycloneDDS middleware within the ROS2 framework. The Raspberry Pi can be accessed remotely through SSH during experimentation.

The physical platform and its onboard hardware configuration are shown in Fig. 2.

The platform is powered by two independent 3-cell LiPo battery packs. One battery supplies the propulsion system, including the motor and steering actuators, while the second supplies the onboard electronics. The onboard electronics are powered through the voltage output of the Yahboom expansion board.

C. SLAM and ROS2 Framework

The robotic platform operates under ROS2 Jazzy Jalisco, which provides the communication infrastructure for sensor data, navigation, and online mapping. Mapping is implemented using the asynchronous mode of the `slam_toolbox` package to generate a two-dimensional occupancy grid from LiDAR observations during vehicle operation. The grid resolution is set to 0.05 m per cell.

The SLAM framework receives LiDAR scans and odometry information derived from the incremental encoder and the IMU-based heading estimate. The planar position is updated through discrete-time dead reckoning:

$$x(t + \Delta t) = x(t) + v(t) \cos\left(\theta(t)\right) \Delta t, \tag{2}$$

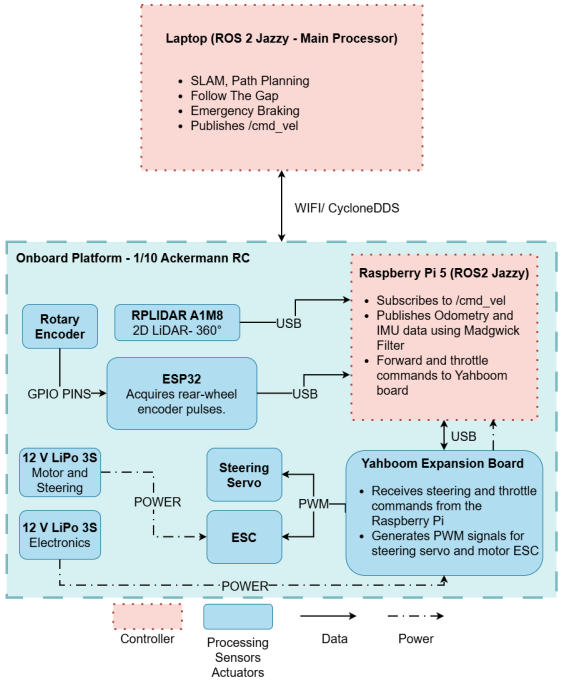


Fig. 1. System architecture of the Ackermann-steered platform, showing the onboard sensors, Raspberry Pi, ESP32-based incremental encoder acquisition, Yahboom expansion board, actuators, and laptop-based ROS2 processing. Data communication between the onboard platform and laptop uses Wi-Fi with CycloneDDS; power connections are indicated separately.

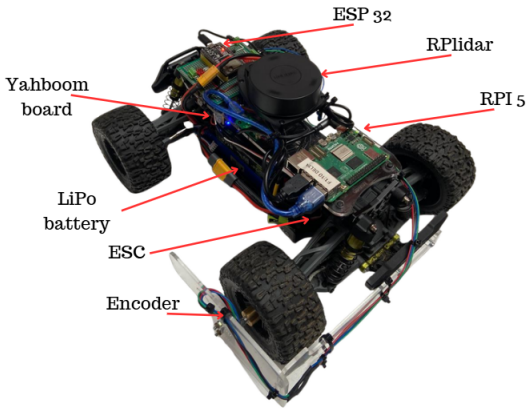


Fig. 2. Experimental Ackermann-steered RC platform and onboard hardware configuration.

$$y(t+\Delta t) = y(t) + v(t)\sin\left(\theta(t)\right)\Delta t, \tag{3}$$

where $x(t)$ and $y(t)$ are the estimated planar coordinates, $v(t)$ is the encoder-derived longitudinal velocity, $\theta(t)$ is the heading estimated by the Madgwick filter, and Δt is the elapsed time between consecutive odometry updates. These equations integrate the estimated linear velocity along the estimated heading. The resulting position estimate is subject to accumulated dead-reckoning error. LiDAR scan matching

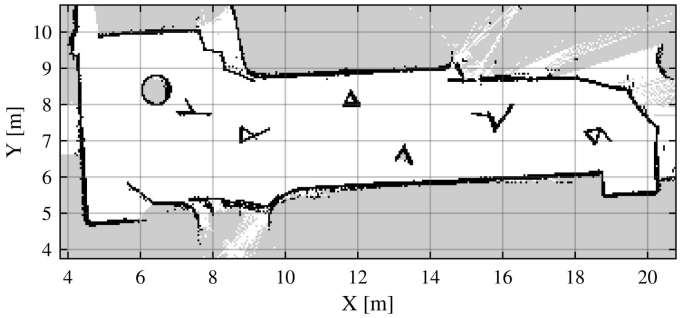


Fig. 3. Occupancy-grid map generated online by `slam_toolbox` during the single physical FTG navigation run.

within `slam_toolbox` uses the observed geometry to estimate the robot pose and construct the occupancy-grid map.

As shown in Fig. 1, the Raspberry Pi is responsible for onboard sensing and low-level communication, whereas SLAM and high-level navigation processing are executed on the external laptop. The two systems exchange ROS2 messages over Wi-Fi using CycloneDDS. Computational performance was not quantitatively profiled during the reported experiment. Consequently, no conclusions are drawn regarding CPU utilization, memory consumption, communication latency, or mapping update rate.

D. Experimental Environment and Procedure

The system was evaluated on an indoor test track at the EIA University facilities in Envigado, Colombia. The corridor-like environment measures approximately 15 m in length and 4 m in width and is bounded by rigid side walls. Existing doors and junctions were sealed using plastic panels and solid objects to create a confined test area. Multiple static obstacles were distributed throughout the track.

During the experiment, the vehicle used FTG for reactive navigation while `slam_toolbox` generated the occupancy-grid map online from LiDAR observations and odometry information. The resulting map is presented in Fig. 3. It represents the map generated during the reported physical run, rather than a reference map prepared before the experiment.

The experimental evaluation consisted of a single physical run. The reported traveled distance of 38.601 m corresponds to this run and is not an average across repeated trials. Exploration was terminated manually by the operator. Therefore, the experiment characterizes the observed behavior of the integrated navigation and mapping system during this traversal; it does not evaluate autonomous exploration termination or establish repeatability across runs.

III. Results and Analysis

This section presents the quantitative and qualitative results obtained during a single physical run of the proposed FTG-SLAM framework. The analysis reports the observed navigation metrics and examines the final occupancy-grid map and estimated vehicle trajectory. Since the experiment

was terminated manually and no repeated trials or ground-truth measurements were available, the results are interpreted as an initial demonstration of system operation under the tested conditions rather than as a comprehensive evaluation of repeatability, safety, or mapping accuracy.

Table I summarizes the key performance metrics obtained during the autonomous exploration process.

TABLE I
REACTIVE NAVIGATION PERFORMANCE METRICS FOR A SINGLE AUTONOMOUS EXPLORATION RUN

Metric	Value
Emergency-Braking Activations	0
Distance Traveled [m]	38.601
Mean Forward Velocity [m/s]	0.507
Std. Forward Velocity [m/s]	0.116
Mean Angular Velocity [rad/s]	0.023
Std. Angular Velocity [rad/s]	0.556

During the recorded run, the vehicle traveled 38.601 m while sensor data were acquired and the occupancy-grid map was updated. No emergency-braking activations were recorded. This observation indicates that the emergency-braking mechanism was not triggered during the experiment; however, without measurements of obstacle clearance, time-to-collision, or near-collision events, it does not establish a quantitative safety margin or demonstrate collision avoidance. Moreover, because the evaluation consisted of a single run, the results do not establish the repeatability or robustness of the navigation strategy.

The mean forward velocity was 0.507 m/s, with a standard deviation of 0.116 m/s. These values summarize the longitudinal motion observed during the run. Although the mean provides an indication of the operating speed, the aggregate statistics alone do not characterize transient speed variations or establish the effect of the selected speed on mapping quality. The velocity results should therefore be interpreted descriptively, rather than as evidence of improved SLAM convergence or enhanced obstacle-avoidance performance.

The mean angular velocity was 0.023 rad/s, while its standard deviation was 0.556 rad/s. The near-zero mean indicates that the signed angular velocity averaged over the recorded data was small, whereas the larger standard deviation reflects variability in angular motion during the run. These statistics are consistent with changes in heading as the vehicle navigated the environment, although they do not provide a direct assessment of steering accuracy, obstacle clearance, or navigation safety.

Figure 4 presents the final occupancy-grid map generated during the run together with the estimated vehicle trajectory. The figure provides a qualitative view of the environment representation produced by SLAM and the path followed by the Ackermann-steered platform under reactive navigation based on FTG.

The final map and estimated trajectory provide a qualitative illustration of the regions represented during the experiment.

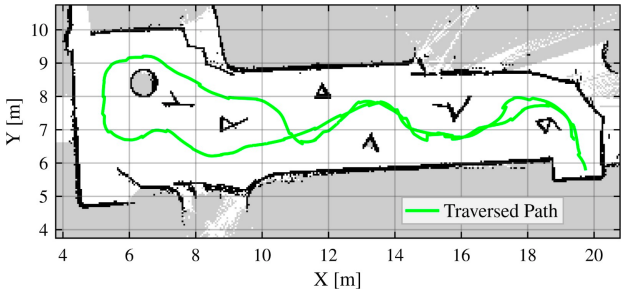


Fig. 4. Final occupancy-grid map and estimated vehicle trajectory during the single FTG-based navigation run.

The figure allows the spatial correspondence between the recorded trajectory and the resulting occupancy-grid representation to be inspected visually. However, because no reference map or quantitative coverage metric was available, the completeness of the explored area and the geometric accuracy of the generated map cannot be established from this experiment alone.

The results demonstrate the operation of the integrated FTG reactive navigation and SLAM mapping framework on a physical Ackermann-steered platform. During the recorded run, the vehicle traveled 38.601 m without triggering the emergency brake mechanism, while an occupancy-grid map was generated online. The final map and trajectory provide qualitative evidence of the system's operation in the tested environment. Nevertheless, the single-run evaluation, manual termination of the experiment, and absence of quantitative safety, coverage, mapping-accuracy, and computational-performance measurements limit the conclusions that can be drawn. Repeated trials and quantitative evaluation against appropriate reference measurements are required to assess repeatability, safety, completeness of exploration, and mapping quality. The generated map may provide a basis for future map-based navigation experiments, but its suitability for that purpose has not been validated here.

IV. CONCLUSIONS

This work presented the implementation and experimental validation of a reactive autonomous exploration framework for occupancy-grid map generation in previously unknown environments. The proposed approach combines Follow-the-Gap (FTG) reactive navigation with LiDAR-based SLAM under a ROS2 architecture deployed on a real Ackermann-steered robotic platform. The experimental results demonstrated that the vehicle was capable of safely traversing the environment while continuously generating a geometrically coherent map without requiring prior environmental knowledge, predefined trajectories, or external intervention. Furthermore, the absence of emergency braking activations during the experiments validated the effectiveness of the proposed reactive navigation strategy in maintaining safe obstacle clearance throughout the exploration process.

The obtained results also showed that conservative velocity management contributes positively to both navigation safety

and mapping quality. By limiting the operating speed, the system achieved stable reactive behavior while simultaneously improving scan registration conditions for the SLAM framework. Although the employed odometry estimation was based on relatively simple wheel encoder and inertial sensing, the integration with `slam_toolbox` enabled sufficiently consistent pose estimation for reliable occupancy-grid generation. Nevertheless, the proposed framework still presents several limitations. As a purely reactive method, FTG does not explicitly optimize exploration efficiency or guarantee complete coverage in highly complex environments. Additionally, mapping performance remains dependent on the quality of LiDAR observations, environmental geometric richness, and the short-term consistency of the odometry estimation.

Future work may focus on integrating frontier-based exploration or active SLAM techniques to improve exploration completeness and reduce redundant traversals. Additional research could also investigate the incorporation of sensor fusion approaches based on Kalman filtering or visual-inertial odometry to improve localization robustness under challenging operating conditions. Finally, extending the framework toward fully onboard computation and evaluating its performance in larger-scale or dynamic environments would further contribute to the development of lightweight autonomous mapping systems for real-world robotic applications.

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